
HOMEWORK DAY 6 – *Derivatives of inverse trig functions §6.6*

1. *Derivatives of inverse trig functions.* Find the derivatives of the following functions. Simplify if possible.

(a) §6.6: 22.

(b) §6.6: 23.

(c) §6.6: 24.

(d) §6.6: 25.

(e) §6.6: 27.

(f) §6.6: 35.

2. *Limits of inverse trig functions.* Find the following limits

(a) §6.6: 45.

(b) §6.6: 46.

(c) §6.6: 47.

(d) §6.6: 48.

3. *Integrals.* Evaluate the following integrals

(a) §6.6: 69. $\int \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}} dx = \int e^u du = e^u + C = e^{\sin^{-1} x} + C$ where $u = \sin^{-1} x, du = \frac{1}{\sqrt{1-x^2}} dx$

(b) §6.6: 71.

(c) §6.6: 72.

(d) §6.6: 73.

(e) §6.6: 74.

HOMEWORK DAY 7 – *Hyperbolic functions §6.7*

4. Find the derivative of the following functions. Simplify if possible.

(a) §6.7: 36

(b) §6.7: 38

(c) §6.7: 39

(d) §6.7: 40

(e) §6.7: 41

(f) §6.7: 42

5. Prove that $\cosh^2(x) - \sinh^2(x) = 1$

6. Sketch the graph of $y = \cosh x$. In your sketch, indicate all the solutions to $\cosh x = 2$. Then solve $\cosh x = 2$ to find the exact values. (Hint: use the definition $\cosh x = (e^x + e^{-x})/2$.)

7. Use the definition $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ to answer the questions below.

(a) Is $\tanh(x)$ odd or even? Show your work. (Use the definition of an odd/even function.)

(b) Find $\lim_{x \rightarrow \infty} \tanh(x)$

(c) Find $\lim_{x \rightarrow -\infty} \tanh(x)$

(d) State all intercepts and all asymptotes of $y = \tanh(x)$:

(e) Show that $f(x) = \tanh(x)$ is always increasing.

(f) Use the information above to sketch a graph of $y = \tanh(x)$.

8. If a water wave with length L moves with velocity v in a body of water with depth d , then

$$v = \sqrt{\frac{gL}{2\pi}} \tanh\left(\frac{2\pi d}{L}\right)$$

where g is acceleration due to gravity. Explain why the approximation

$$v \approx \sqrt{\frac{gL}{2\pi}}$$

is appropriate in deep water.

HOMEWORK DAY 8 – *L'Hôpital's rule §6.8*

9. §6.8: 32. $\lim_{x \rightarrow \infty} \frac{(\ln x)^2}{x} \stackrel{(\frac{\infty}{\infty})}{=} \lim_{x \rightarrow \infty} \frac{2(\ln x)/x}{1} = \lim_{x \rightarrow \infty} \frac{2 \ln x}{x} \stackrel{(\frac{\infty}{\infty})}{=} \lim_{x \rightarrow \infty} \frac{2/x}{1} = 0$

10. §6.8: 10.

11. §6.8: 11.

12. §6.8: 13.

13. §6.8: 14.

14. §6.8: 15.

15. §6.8: 24.

16. §6.8: 25.

17. A drug response curve describes the level of medication in the bloodstream after a drug is administered. A surge function $S(t) = At^pe^{-kt}$ is often used to model the response curve, reflecting an initial surge in the drug level and then a more gradual decline. The parameters $A, k > 0$ are positive and depend on the particular drug.

Consider the drug response curve $S(t) = At^2e^{-kt}$, $t \geq 0$.

- (a) Find all intercepts.

(b) Find $\lim_{t \rightarrow \infty} S(t) =$

(c) Find all local minimums and maximums of $S(t)$ for $t \geq 0$. Justify your answer.

(d) Use your results to sketch a graph of the function.

(e) What is the maximal drug response and when does it occur? Label that point in your graph.

18. Let $f(x) = \frac{\sin x}{x}$.

(a) Determine whether f is even or odd.

(b) State the domain and find all intercepts.

(c) Find $\lim_{x \rightarrow \infty} f(x) =$

(d) Find $\lim_{x \rightarrow 0} f(x) =$

(e) Use the above to sketch a clearly labeled graph of f for $x \in [-5\pi, 5\pi]$.

19. Find the following limits, if they exist.

(a) $\lim_{x \rightarrow 0^+} x \ln x$

(b) $\lim_{x \rightarrow -\infty} x e^x$

(c) $\lim_{t \rightarrow -\infty} A t^2 e^{-kt}$, where $A, k > 0$

(d) $\lim_{x \rightarrow \infty} \frac{\ln(x)}{\sqrt{x}}$

(e) $\lim_{x \rightarrow 1} \frac{\ln(x)}{\sqrt{x}}$

$$(f) \quad \lim_{x \rightarrow 0^+} \frac{\ln(x)}{\sqrt{x}}$$

$$(g) \quad \lim_{x \rightarrow \infty} \left(\ln(x) - \ln(x+1) \right)$$

$$(h) \quad \lim_{n \rightarrow \infty} \left(1 + \frac{r}{n} \right)^n, \quad r \in \mathbb{R}$$